

AI-Powered Customer Support Ticketing System

Project Documentation

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APSCHE – Salesforce Certified Administrator with AI Agentforce

Academic Year: 2025–2026

Declaration

I, **Y. Naga Mounika** (Roll No. **24A35A4608**), hereby declare that the project titled "**AI-Powered Customer Support Ticketing System**" is my original work completed as part of the **APSCHE – Salesforce Certified Administrator with AI Agentforce Internship**. The work presented in this report has not been submitted elsewhere for any other certification or academic purpose.

Acknowledgement

I sincerely express my gratitude to the **Andhra Pradesh State Council of Higher Education (APSCHE)** for providing me with the opportunity to participate in the **Salesforce Certified Administrator with AI Agentforce Internship**. I also thank **Salesforce, SmartBridge**, and the **Trailhead learning platform** for providing valuable learning resources and hands-on experience throughout the internship. Finally, I extend my heartfelt thanks to my family and friends for their continuous encouragement and support during the successful completion of this project.

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AI-Powered Customer Support Ticketing System

User Story:

The **AI-Powered Customer Support Ticketing System** is designed to help support teams efficiently manage customer service requests through Salesforce. As a **Support Agent**, I want to quickly access customer information, create and update support cases, classify issues based on their type, and receive automated priority assignments and SLA due dates. This automation reduces manual effort, ensures faster response times, and helps deliver consistent, high-quality customer support.

As a **Support Manager and Administrator**, I want to monitor support operations using reports and dashboards, manage users through profiles and permission sets, and ensure secure, role-based access to system resources. I also want the system to capture AI feedback and provide AI-assisted suggestions for case handling, enabling continuous process improvement and creating a scalable, intelligent customer support solution built on Salesforce.

1. INTRODUCTION

1.1 Project Overview :

The **AI-Powered Customer Support Ticketing System** is a Salesforce-based customer support management solution designed to streamline and automate the process of handling customer service requests. The system centralizes customer information using Salesforce standard objects such as **Account, Contact, and Case**, while extending functionality with custom fields and automation to improve support efficiency. It enables support teams to create, track, prioritize, and resolve customer issues through a structured and secure workflow.

The project leverages Salesforce **Flows, Validation Rules, Profiles, Permission Sets, Reports, and Queues** to automate case classification, priority assignment, SLA due date calculation, and role-based access control. It also incorporates AI-oriented features such as **AI Suggested Reply, AI Suggested Response, and AI Feedback** to assist support agents and support continuous improvement. The project user story, objectives, and system requirements are presented in the following sections of this document.

1.2 Purpose :

The purpose of this project is to:

- Streamline customer support operations by centralizing customer and case management in Salesforce.
- Automate case classification, priority assignment, and SLA due date calculation using Salesforce Flows.
- Improve the accuracy and efficiency of support processes by minimizing manual intervention through automation.
- Implement role-based access control using Profiles and Permission Sets to ensure secure access to customer data.
- Generate reports to monitor case status, issue types, priority distribution, and overall support performance.
- Generate reports to monitor case status, issue types, priority distribution, and overall support performance.
- Provide a scalable and maintainable customer support solution using Salesforce standard features and low-code automation.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Support Agent	Manage customer support cases efficiently	Case prioritization and assignment are performed manually	There is no automated workflow for classifying and routing cases	Overloaded and less productive
PS-2	Support Manager	Monitor support performance and SLA compliance	It is difficult to track case status and team performance	Reports and analytics are not centralized	Frustrated and unable to make timely decisions
PS-3	Customer	Receive quick and accurate resolutions for support requests	My issues are delayed or not prioritized correctly	The support process lacks automation and intelligent assistance	Dissatisfied and undervalued

Problem Statements – AI-Powered Customer Support Ticketing System					
PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	 Support Agent	 Manage customer support cases efficiently	 Case prioritization and assignment are done manually	 There is no automated workflow for classifying and routing cases	 Overloaded and less productive
PS-2	 Support Manager	 Monitor case status and SLA compliance and team performance	 Reports are scattered and difficult to analyze	 There is no centralized reporting and analytics system	 Frustrated and unable to make timely decisions
PS-3	 Customer	 Receive quick and accurate support	 Support requests are delayed or incorrectly prioritized	 The support process lacks automation and intelligent assistance	 Dissatisfied and undervalued

2.2 Empathy Map Canvas

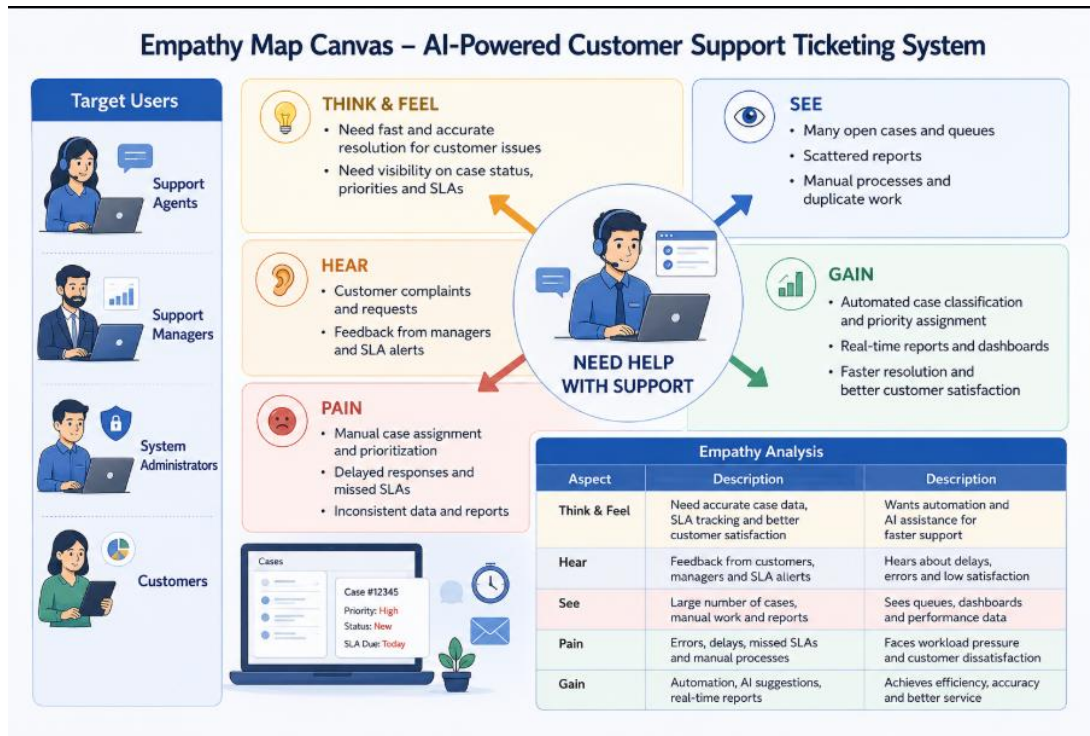
Target Users

- Support Agents
- Support Managers
- System Administrators
- Customers

Empathy Analysis

Aspect	Description
Think & Feel	Need quick and accurate customer support.
Hear	Customer complaints and manager feedback.
See	Many support tickets and manual work.

Say & Do	Create, Update, and Resolve support cases
Pain	Delayed responses and manual case handling.
Gain	Faster support through automation and AI assistance.



2.3 Brainstorming

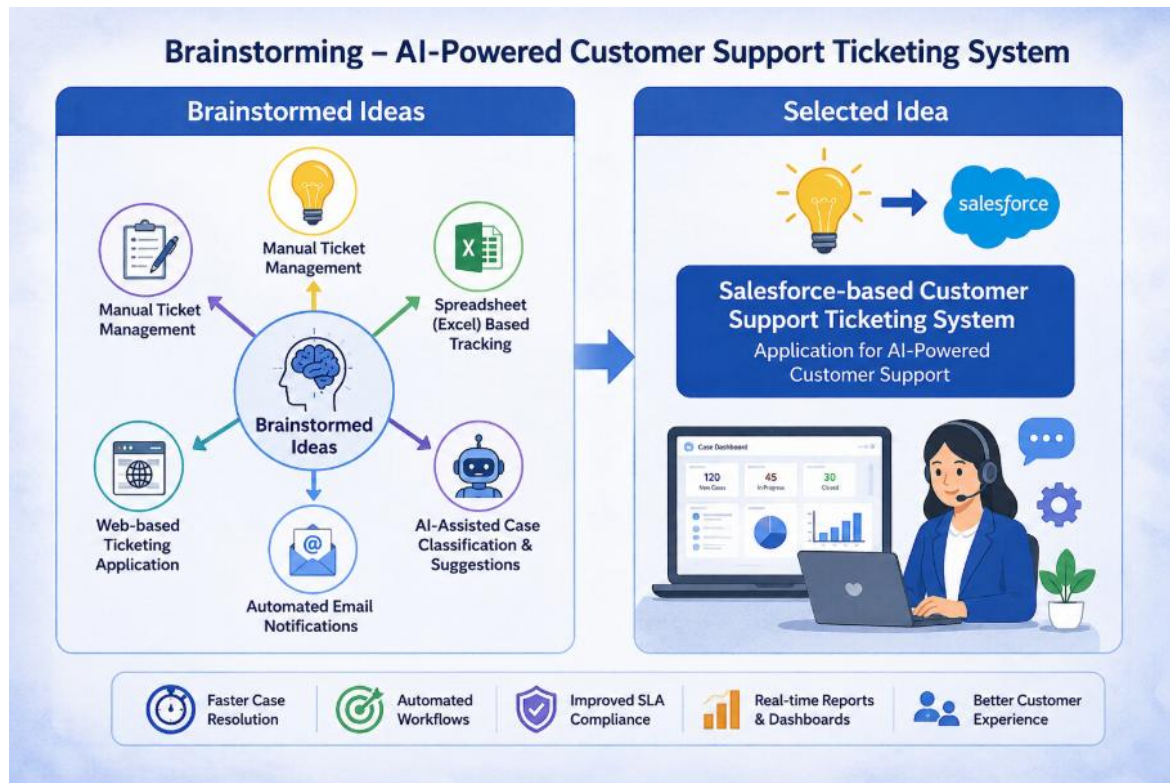
Brainstormed Ideas

- Manual ticket management
- Excel-based tracking
- Web-based support system

- Salesforce support system
- AI-powered ticketing system

Selected Idea:

Salesforce-based CRM Application for **AI-Powered Customer Support Ticketing System**



3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Journey Stages:

- Login to Salesforce
- Create or View Support Case
- Enter Customer and Case Details

- AI-Based Case Classification
- Priority & SLA Assignment
- Resolve the Case
- View Reports & Dashboards
- Monitor Performance



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Login and Authentication
FR-2	Customer and Case Management
FR-3	AI-Based Case Classification

FR-4	Priority and SLA Automation
FR-5	Reports and Dashboards
FR-6	AI Feedback Collection

Non-Functional Requirements NFR- 1: Usability

The system should be simple and easy for support agents, managers, administrators, and customers to use.

Requirements :

- Simple and intuitive user interface
- Clear navigation menus
- Easy-to-understand forms
- User-friendly dashboards

Example:

A support agent should be able to log in, create a case, and update its status easily using the Salesforce Lightning interface.

NFR-2: Security

The system should protect customer and case information from unauthorized access.

Requirements:

- Role-based access control
- Secure login authentication
- Object and field-level security

- Permission Sets and Profiles

Examples:

Only authorized users can edit cases, while managers have additional reporting access.

NFR-3: Reliability

The system should store customer and case information accurately without data loss.

Requirements:

- Automatic data saving
- Reliable cloud storage
- Error handling
- Consistent system performance

Example :

Cases remain safely stored even if the user logs out unexpectedly.

NFR-4: Performance

The system should process cases and reports quickly.

Requirements:

- Fast page loading
- Quick report generation
- Efficient report generation
- Smooth dashboard performance

Example:

Reports should open within a few seconds, even with many case records.

NFR-5: Availability

The system should be available whenever support teams need it.

Requirements:

- 24/7 system availability
- Minimal downtime
- Cloud-based access
- Reliable Salesforce platform

Example:

Support agents can access customer cases anytime from the Salesforce cloud.

NFR-6: Scalability

The system should support future growth without affecting performance.

Requirements:

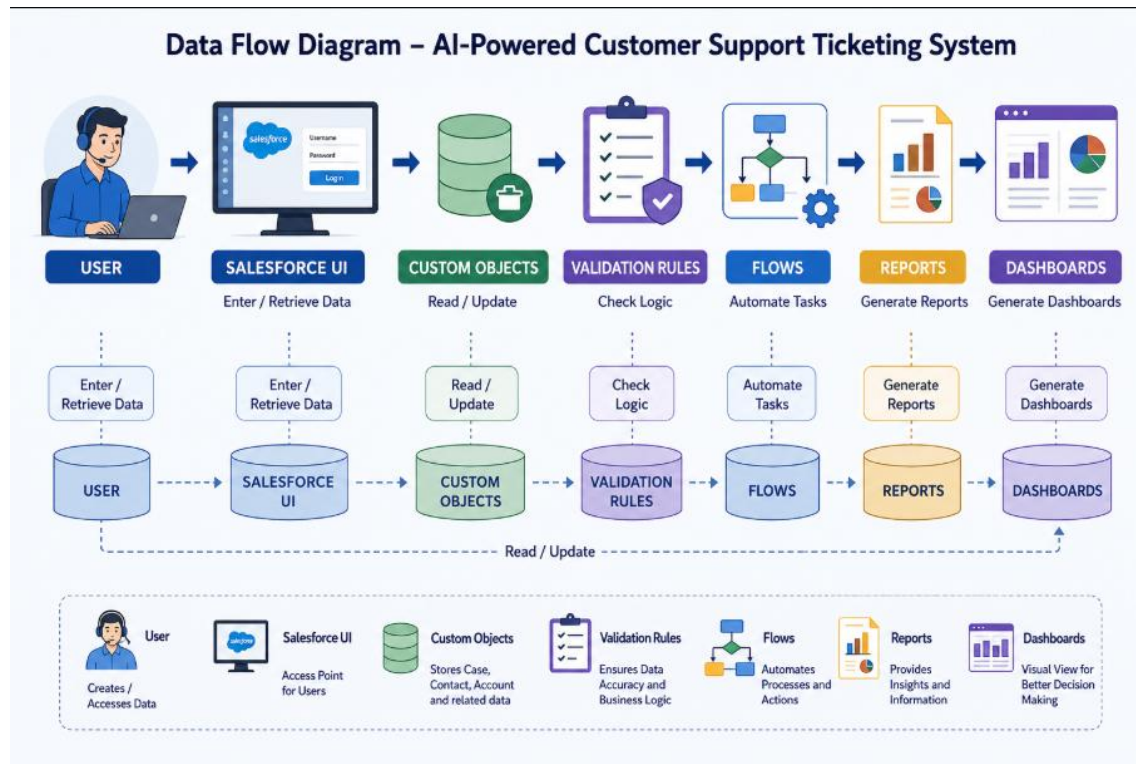
- Support more users
- Handle increasing case records
- Expand storage
- Maintain system speed

Example:

The system can manage thousands of customer support cases as the organization grows.

3.3 Data Flow Diagram

Customer/User → Salesforce Login → Case Object → Validation Rules → Record-Triggered Flows → AI Feedback → Reports → Dashboards.



3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning Experience
Database	Salesforce Cloud
Automation	Salesforce Flows
Backend	Salesforce Platform
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The AI-Powered Customer Support Ticketing System replaces manual case handling with automated case management. Salesforce Flows automate case classification, priority assignment, and SLA tracking, while reports and dashboards provide real-time insights. Role-based security ensures controlled access, improving efficiency, response time, and customer satisfaction.



4.2 Proposed Solution

Functional Requirements

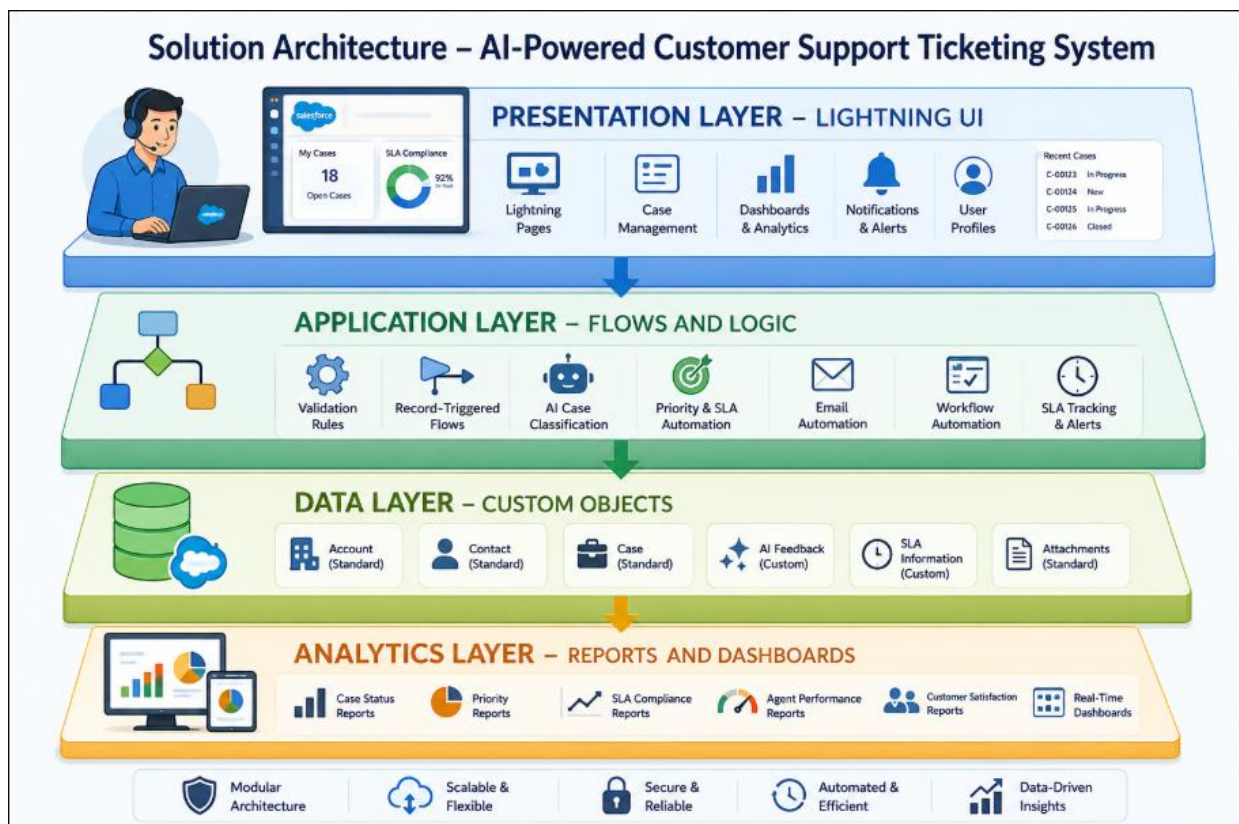
Parameter	Description
User Login & Authentication	Allows authorized users to securely access the Salesforce application using their credentials..
Customer & Contact Management	Enables users to create, view, and manage customer account and contact information.
Case Management	Allows support agents to create, update, assign, and resolve customer support cases.
AI-Based Case Classification	Automatically categorizes support cases based on the selected issue type using Salesforce Flows.
Priority Assignment	Automatically assigns case priority based on issue type and business rules.
SLA Management	Calculates and tracks SLA Due Dates to ensure timely case resolution.
Validation Rules	Ensures mandatory fields and valid data are entered before saving a case.
Custom Fields & Page Layouts	Displays AI-related fields, SLA information, and case details through customized page layouts.
Workflow & Flow Automation	Automates case processing, priority assignment, and SLA updates using Record-Triggered Flows.
AI Feedback Management	Captures agent feedback on AI suggestions to improve future recommendations.
Reports	Generates reports to monitor case status, priority distribution, SLA compliance, and resolution time.
Dashboards	Provides visual insights into support performance using charts and dashboards.
Role-Based Security	Controls access using Profiles, Permission Sets, Object Permissions, and Field-Level Security.
Cloud Storage	Securely stores all customer and support case data in the Salesforce Cloud.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning Experience (User Interface)
2. Application Layer – Validation Rules, Record-Triggered Flows, Business Logic
3. Data Layer – Standard Objects (Account, Contact, Case) and Custom Object (AI Feedback)
4. Security Layer – Profiles, Permission Sets, Field-Level Security, Object Permissions
5. Analytics Layer – Reports and Dashboards

The AI-Powered Customer Support Ticketing System follows a layered Salesforce architecture. Users interact with the Lightning interface to manage customer accounts and support cases. Validation Rules and Record-Triggered Flows automate case classification, priority assignment, and SLA tracking. Customer and case information is securely stored using Salesforce standard and custom objects. Profiles and Permission Sets enforce role-based access control, while Reports and Dashboards provide real-time insights into support operations and overall system performance



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- The project followed an Agile-based development approach with planned milestones and iterative implementation.
- A Salesforce Developer account was created to set up the development environment.
- Standard objects such as **Account, Contact, and Case** and the custom object **AI Feedback** were configured.
- Custom fields, page layouts, validation rules, and field history tracking were implemented.
- Record-triggered Flows were developed to automate **case classification, priority assignment, and SLA tracking**.
- Profiles, Permission Sets, and user accounts were created to implement role-based access control.
- Reports and dashboards were developed to monitor case performance and support operations.
- System testing was performed after every sprint to validate functionality.
- The project was completed according to the planned schedule.

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member s
Sprint-1	Environment Setup	US-1	Configure Salesforce Developer Org and enable required features	2	High	Self
Sprint-1	Data Modeling	US-2	Configure Account, Contact, Case, and AI Feedback objects	5	High	Self
Sprint-2	Data Modeling	US-3	Create custom fields, relationships, and page layouts.	5	High	Self

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Security	US-4	Configure Profiles, Permission Sets, and Users.	5	High	Self
Sprint-3	Automation	US-5	Implement Validation Rules.	3	High	Self
Sprint-3	Automation	US-6	Develop Record-Triggered Flows for Case Automation	8	High	Self
Sprint-4	Reporting	US-7	Create Reports for Case Monitoring	4	High	Self
Sprint-4	Dashboards	US-8	Develop Dashboards for Performance Analysis	3	Medium	Self
Sprint-5	Testing	US-9	Test Flows, Security, Reports, and User Access.	5	High	Self
Sprint-5	Documentation	US-10	Prepare Project Documentation and Final Review.	3	Medium	Self

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	7	2 Days	24 Jun 2026	25 Jun 2026	7	25 Jun 2026
Sprint-2	10	3 Days	26 Jun 2026	28 Jun 2026	10	28 Jun 2026
Sprint-3	11	3 Days	29 Jun 2026	01 Jul 2026	11	01 Jul 2026
Sprint-4	7	2 Days	02 Jul 2026	03 Jul 2026	7	03 Jul 2026
Sprint-5	8	2 Days	04 Jul 2026	05 Jul 2026	8	05 Jul 2026

6. Project Development Phase:

Project Flow

- **Milestone 1:** Salesforce Developer Environment Setup
- **Milestone 2:** Configure Standard & Custom Objects
- **Milestone 3:** Create Custom Fields & Relationships
- **Milestone 4:** Configure Page Layouts
- **Milestone 5:** Implement Validation Rules
- **Milestone 6:** Configure Profiles, Permission Sets & Users
- **Milestone 7:** Develop Record-Triggered Flows
- **Milestone 8:** Configure Field History Tracking
- **Milestone 9:** Create Reports
- **Milestone 10:** Build Dashboards
- **Milestone 11:** System Testing
- **Milestone 12:** Documentation & Final Review

What you'll learn

- Salesforce Administration
- Standard and Custom Objects
- Custom Fields and Relationships
- Page Layouts
- Validation Rules
- Profiles and Permission Sets
- User Management
- Record-Triggered Flows
- Field History Tracking
- Reports and Dashboards
- Role-Based Security
- AI-Based Customer Support Automation

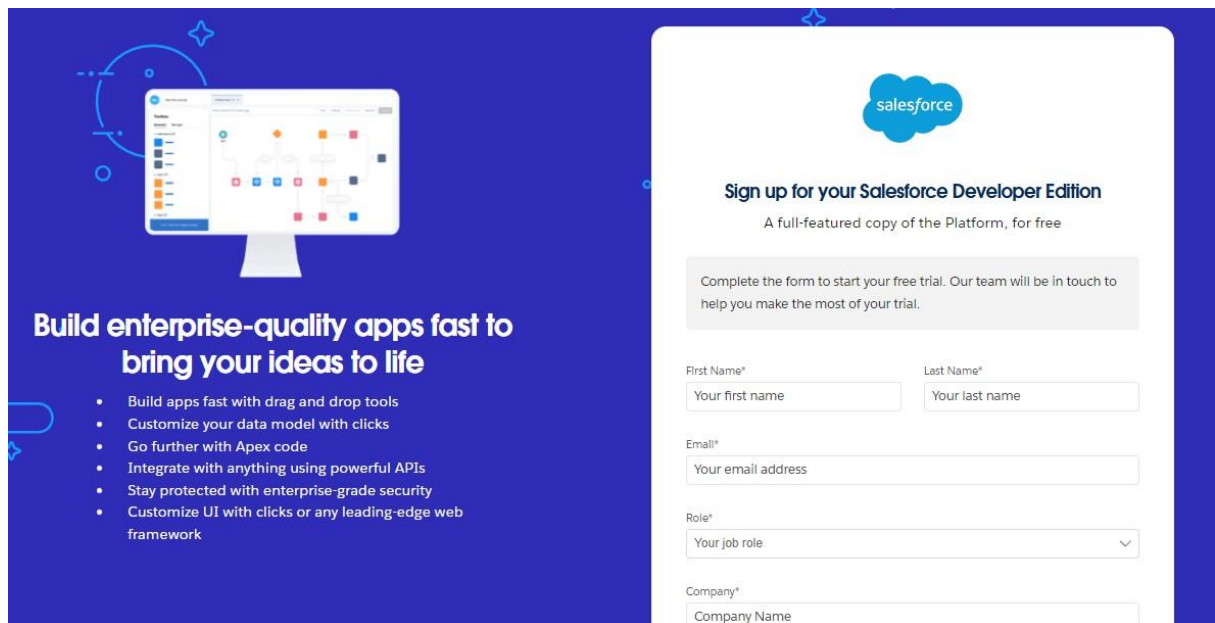
Implementation

Milestone 1 - Salesforce developer Environment Setup:

Activity 1: Creating Developer Account:

A Salesforce Developer Edition environment was created and configured to develop the **AI-Powered Customer Support Ticketing System**. This environment provides access to Salesforce standard objects, automation tools, reports, dashboards, security features, and Flow Builder required for project implementation.

1. Go to <https://developer.salesforce.com/signup>
2. On the sign up form, enter the following details :



Build enterprise-quality apps fast to bring your ideas to life

- Build apps fast with drag and drop tools
- Customize your data model with clicks
- Go further with Apex code
- Integrate with anything using powerful APIs
- Stay protected with enterprise-grade security
- Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*
Your first name

Last Name*
Your last name

Email*
Your email address

Role*
Your job role

Company*
Company Name

- 1) First name & Last name
- 2) Email
- 3) Role : Developer
- 4) Company : College or Company Name
- 5) Country : India

6) Postal Code : pin code

7) Username : should be a combination of your name and company

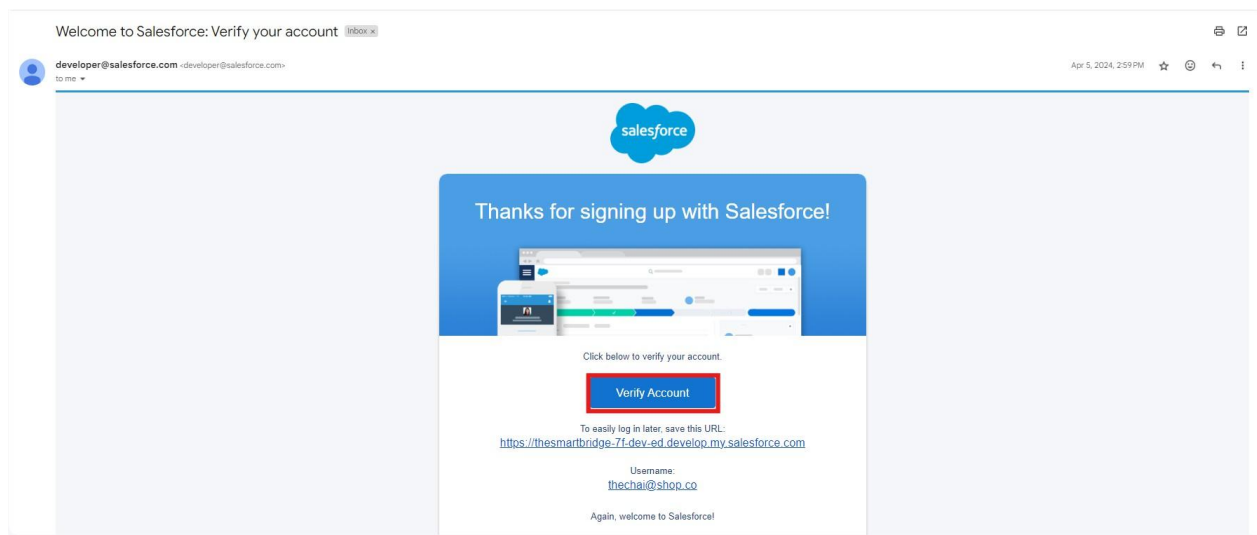
This need not be an actual email id, you can give anything in that format :

username@organization.com

Click on sign me up after filling these.

Activity 2: Account Activation :

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account.
3. Give a password and answer a security question and click on change password.

Change Your Password

Enter a new password for lead@sb.oom.
Make sure to include at least:

- ✓ 8 characters
- ✓ 1 letter
- ✓ 1 number

* New Password
..... Good

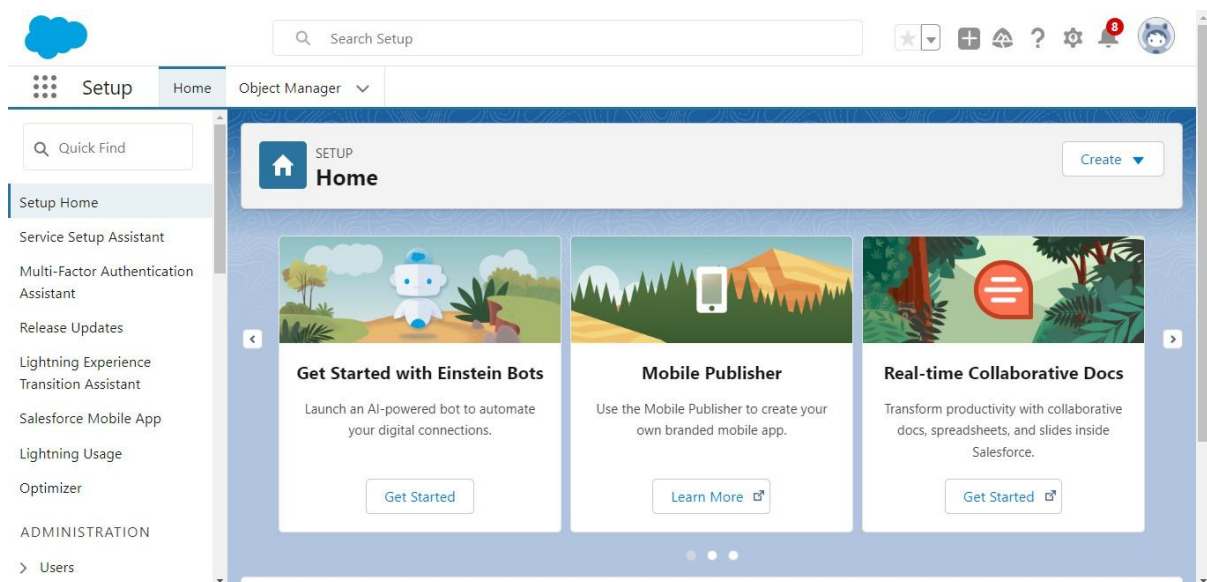
* Confirm New Password
..... Match

Security Question
▼ In what city were you born?

* Answer
asdfghjkl

Change Password

4. Then you will redirect to your salesforce setup page.



Milestone 2 - Object Configuration:

The AI-Powered Customer Support Ticketing System uses Salesforce standard objects along with a custom object to manage customer support operations

Activity 1 : Configure Account Object

The Account object stores customer or organization information

Steps:

1. Go to **Setup**.
2. Open **Object Manager**.
3. Select **Account**.
4. Configure the required fields and page layout.
5. Save the changes.

Purpose:

Stores customer or company details associated with support cases.

Setup

Home

Object Manager



SETUP

Object Manager

S3+ Items, Sorted by Label

LABEL	API NAME	TYPE	DESCRIPTION
Account	Account	Standard Object	

Details

Description

API Name
Account

Custom

Singular Label
Account

Plural Label
Accounts

Enable Reports

Track Activities

Track Field History

Deployment Status

Help Settings
Standard salesforce.com Help Window

Setup

Home

Object Manager

SETUP > OBJECT MANAGER

Account

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Fields & Relationships

33+ Items, Sorted by Field Label

FIELD LABEL

Account Name

Account Number

Account Owner

Account Site

Account Source

Active

FIELD NAME

Name

AccountNumber

OwnerId

Site

AccountSource

Active_c

DATA TYPE

Name

Text(40)

Lookup(User)

Text(80)

Picklist

Picklist

CONTROLLING FIELD

INDEXED

✓

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

Activity 2 : Customer Contact Object

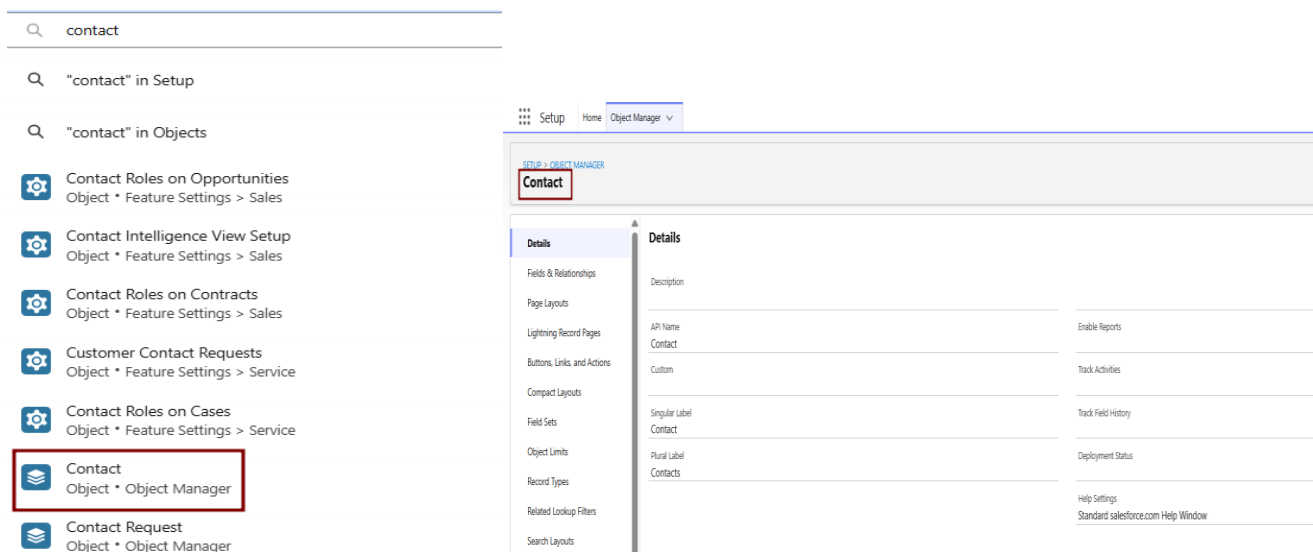
The Contact **object** stores customer contact information

Steps:

1. Go to **Setup**.
2. Open **Object Manager**.
3. Select **Contact**.
4. Configure required fields and page layout.
5. Save the changes

Purpose:

Maintains customer contact details linked to Account and Case records.



Activity 3 : Configure CaseObject

The Case **object** is the primary object used to manage customer support requests

Steps:

1. Go to **Setup**.
2. Open **Object Manager**.
3. Select **Case**.
4. Configure custom fields such as:
 - Issue Type
 - Resolution Time
 - SLA Due Date
 - AI Suggested Reply

- AI Suggested Response
- AI Feedback

5. Enable **Field History Tracking**.
6. Save the configuration.

Purpose:

Stores customer support requests and enables automation for priority assignment, SLA tracking, and case management.

SETUP > OBJECT MANAGER

Case

Details	Fields & Relationships 49 Items, Sorted by Field Label		
Fields & Relationships	Priority	Priority	Picklist
Case Page Layouts	Product	Product__c	Picklist
Case Close Page Layouts	Product	ProductId	Lookup(Product)
Lightning Record Pages	Resolution Time	Resolution_Time__c	Number(5, 0)
Buttons, Links, and Actions	Service Contract	ServiceContractId	Lookup(Service Contract)
Compact Layouts	SLA Due Date	SLA_Due_Date__c	Date/Time
Field Sets	SLA Policy End Time	SLAExitDate	Date/Time
Object Limits	SLA Policy Start Time	SLAStartDate	Date/Time
Record Types	SLA Violation	SLAViolation__c	Picklist
Related Lookup Filters	Status	Status	Picklist
Search Layouts			
List View Button Layout			

Activity -4 : Create AI Feedback Custom Object

The **AI Feedback** custom object stores feedback provided by support agents regarding AI-generated suggestions.

Steps:

1. Go to **Setup**.
2. Open **Object Manager**.
3. Click **Create → Custom Object**.
4. Enter:
 -
 - **Plural Label:** AI Feedbacks
 - **Record Name:** Feedback Name
 - **Data Type:** Text
5. Enable:
 - Allow Reports
 - Track Activities
6. Click **Save**.

Purpose:

Captures feedback on AI-generated responses to support future improvements.

Case Custom Field: **AI Feedback**
[Back to Case Fields](#)

[Validation Rules](#) (0)

Custom Field Definition Detail [Edit](#) [Set Field-Level Security](#) [View Field Accessibility](#) [Where is this used?](#)

Field Information

Field Label	AI Feedback	Object Name	Case
Field Name	AI_Feedback	Data Type	Lookup
API Name	AI_Feedback__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	Yarrabathula Naga Mounika, 6/28/2026, 8:03 AM	Modified By	Yarrabathula Naga Mounika, 6/28/2026, 8:03 AM

Lookup Options

Related To	AI Feedback	Child Relationship Name	Cases
Related List Label	Cases		
Required	☐		
What to do if the lookup record is deleted?	Clear the value of this field.		

Milestone 3 - Case Object Configuration

Activity 1 : Configure Custom Fields

1. Go to **Setup** → **Object Manager** → **Case** → **Fields & Relationships**.
2. Create the required custom fields.

Custom Fields Created

Field Name	Data Type
Issue Type	Picklist
Resolution Time	Number
SLA Due Date	Date/Time
AI Suggested Reply	Long Text Area
AI Suggested Response	Long Text Area
AI Feedback	Lookup (AI Feedback)

SETUP > OBJECT MANAGER

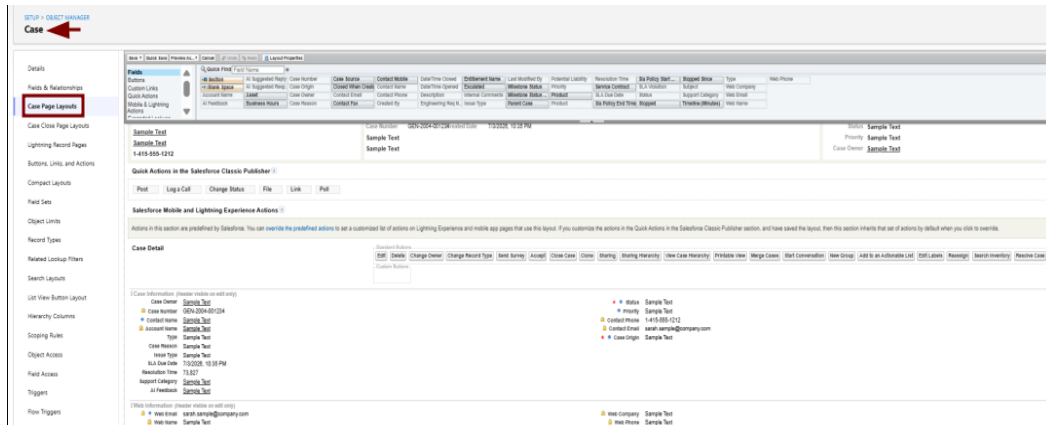
Case

Details	Fields & Relationships 49 Items, Sorted by Field Label
Fields & Relationships	
Case Page Layouts	
Case Close Page Layouts	
Lightning Record Pages	
Buttons, Links, and Actions	
Compact Layouts	

FIELD LABEL	FIELD NAME	DATA TYPE
Account Name	AccountId	Lookup(Account)
AI Feedback	AI_Feedback__c	Lookup(AI Feedback)
AI Suggested Reply	AI_Suggested_Reply__c	Long Text Area(3000)
AI Suggested Response	AI_Suggested_Response__c	Long Text Area(500)
Issue Type	Issue_Type__c	Picklist
Last Modified By	LastModifiedById	Lookup(User)
Milestone Status	MilestoneStatus	Text(30)
Parent Case	ParentId	Lookup(Case)
Potential Liability	PotentialLiability__c	Picklist
Priority	Priority	Picklist
Product	Product__c	Picklist
Product	ProductId	Lookup(Product)
Resolution Time	Resolution_Time__c	Number(5, 0)
Service Contract	ServiceContractId	Lookup(Service Contract)
SLA Due Date	SLA_Due_Date__c	Date/Time

Activity 2: Configure Page Layout

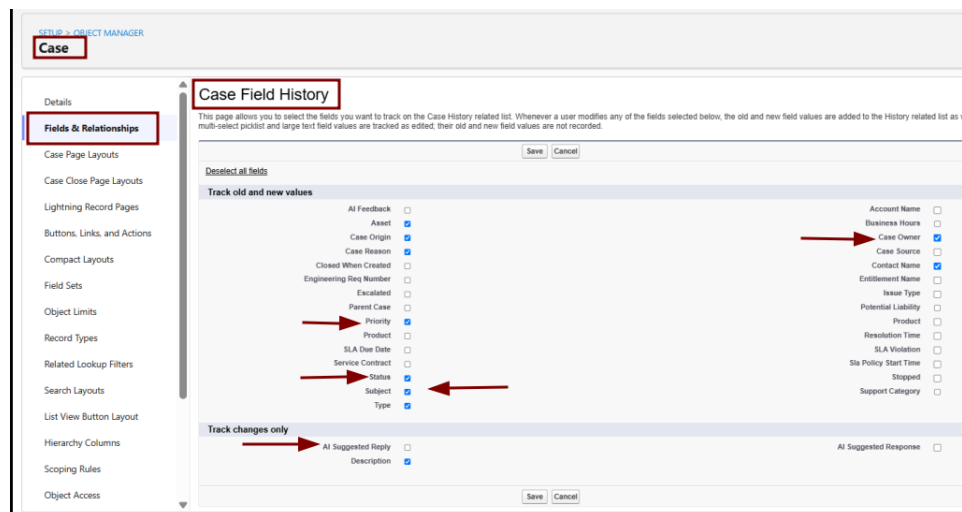
- Open Object Manager → Case.
- Select Page Layouts.
- Add all custom fields to the Case layout.
- Save the layout.



Activity 3: Enable Field History Tracking

1. Open Case → Fields & Relationships.
2. Select Set History Tracking.
3. Enable tracking for important fields such as:

- Status
- Priority
- Subject
- Owner
- AI Suggested Reply



Milestone 4 - Validation Rules

Activity 1 : Create Validation Rules

1. Navigate to **Object Manager** → **Case** → **Validation Rules**.
2. Create validation rules to ensure accurate data entry.

Validation Rules Implemented

- Issue Type is mandatory.
- SLA Due Date is required for High Priority cases

SETUP > OBJECT MANAGER

Case

Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Hierarchy Columns
Scoping Rules
Object Access
Field Access
Triggers
Flow Triggers
Validation Rules
Conditional Field Formatting

Validation Rules
2 Items, Sorted by Rule Name

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE
Require_Issue_Type	Issue Type	Please select an Issue Type.	✓
SLA_Due_Date_Required_For_High_Priority	SLA Due Date	SLA Due Date must be populated for High priority cases.	✓

Case Validation Rule
[Back to Case Validation Rules](#)

Validation Rule Detail Edit Clone

Rule Name	Require_Issue_Type	Active	✓
Error Condition Formula	ISBLANK(TEXT(Issue_Type__c))	Error Location	Issue Type
Error Message	Please select an Issue Type.		
Description	Prevents users from saving a case without selecting an Issue Type.		
Created By	Yarrabathula Naga Mounika, 7/1/2026, 3:58 AM	Modified By	Yarrabathula Naga Mounika, 7/1/2026, 3:58 AM

Edit Clone

Case Validation Rule
[Back to Case Validation Rules](#)

Validation Rule Detail Edit Clone

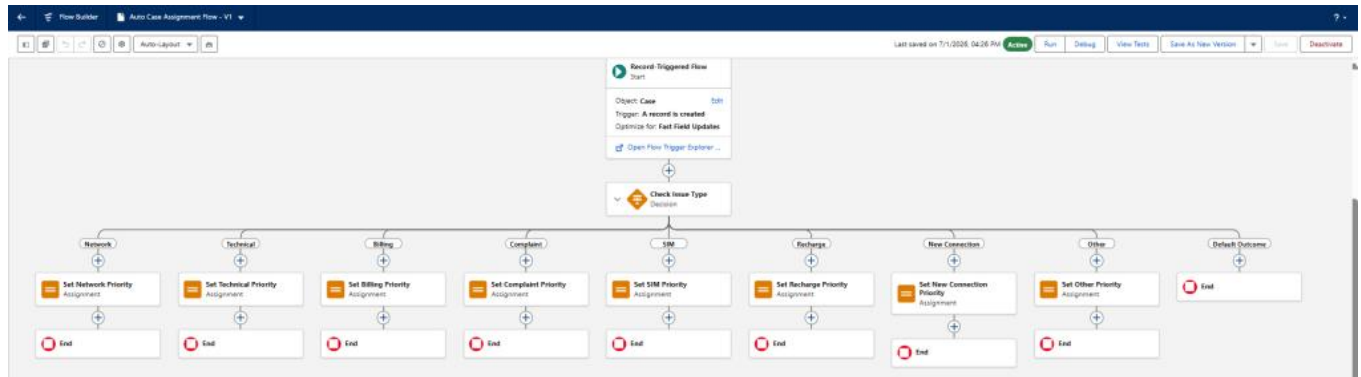
Rule Name	SLA_Due_Date_Required_For_High_Priority	Active	✓
Error Condition Formula	AND(ISPICKVAL(Priority, "High"), ISBLANK(SLA_Due_Date__c))	Error Location	SLA Due Date
Error Message	SLA Due Date must be populated for High priority cases.		
Description	Ensures SLA Due Date is populated for High Priority cases.		
Created By	Yarrabathula Naga Mounika, 6/28/2026, 8:07 AM	Modified By	Yarrabathula Naga Mounika, 6/28/2026, 8:07 AM

Edit Clone

Milestone 5 : Record-Triggered Flow

Activity 1 : Auto Case Assignment Flow

- Go to **Setup** → **Flows**.
- Create a **Record-Triggered Flow**.
- Configure:
 1. Decision element based on **Issue Type**.
 2. Assignment elements for case routing and priority.
- Activate the flow.



This screenshot shows the configuration for the 'Check Issue Type' decision element. The 'Label' field is set to 'Check Issue Type' and the 'API Name' is 'Check_issue_Type'. The 'Description' field is empty. Under the 'Outcomes' section, a list of outcomes is shown on the left, including Network, Technical, Billing, Complaint, SIM, Recharge, New Connection, Other, and Default Outcome. The 'Network' outcome is selected for configuration. The 'Outcome Label' is 'Network' and the 'Outcome API Name' is 'Network'. The 'Condition Requirements to Execute Outcome' are set to 'All Conditions Are Met (AND)'. A red arrow points to the 'Add Condition' button. Below it, a condition is added: 'Resource' is 'Triggering Case > Issue Type', the 'Operator' is 'Equals', and the 'Value' is 'Network'.

OUTCOME ORDER	OUTCOME DETAILS
Network	* Outcome Label: Network * Outcome API Name: Network Condition Requirements to Execute Outcome: All Conditions Are Met (AND) * Resource: Triggering Case > Issue Type X Operator: Equals Value: Network X + Add Condition

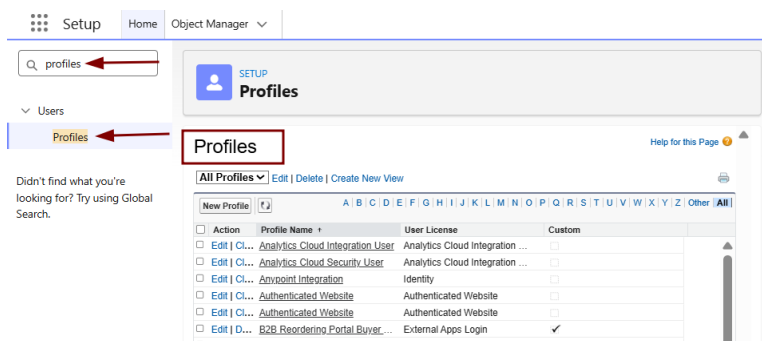
Milestone 6 : Profiles & Permission Sets

Activity 1 : Create Custom Profiles

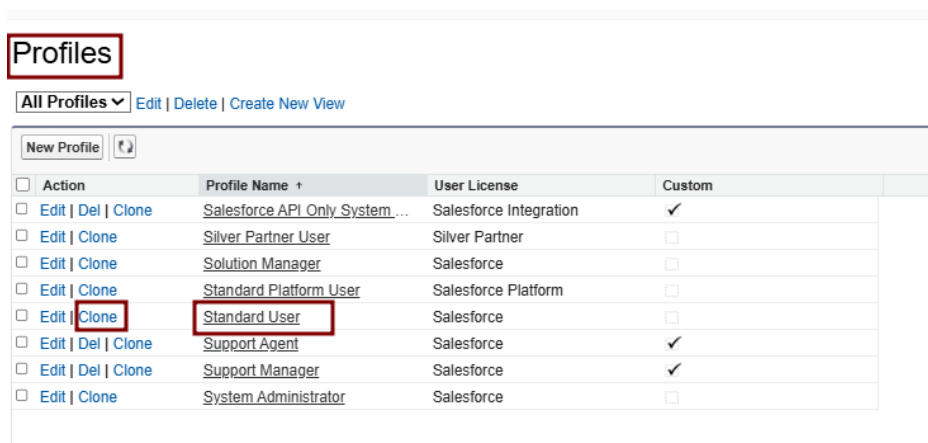
Profiles define the permissions and access levels available to different users within the Salesforce application.

Steps

1. Log in to the Salesforce Developer Org.
2. Navigate to **Setup**.
3. Search for **Profiles** in the Quick Find box.

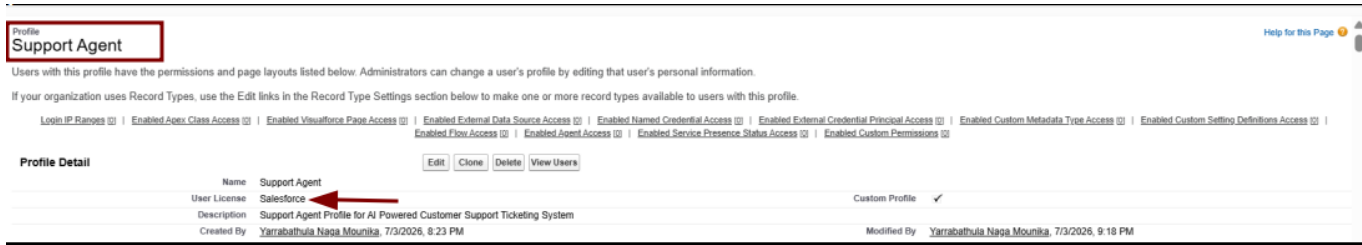


4. Open the **Profiles** page.
5. Clone the **Standard User** profile.



6. Create the following custom profiles:
 - o **Support Agent**
 - o **Support Manager**
7. Save each profile.

Support Agent Profile

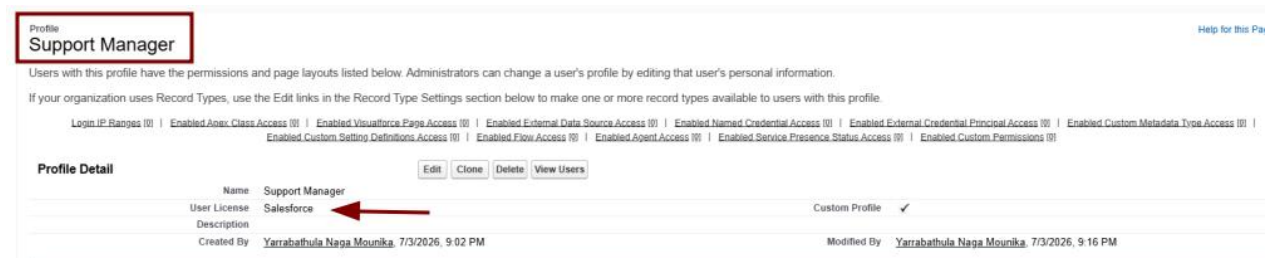


The Support Agent profile was created to provide customer support representatives with access to manage customer cases while restricting administrative functions.

The profile includes:

- Create, Read, Edit, and Delete access for the Case object.
- Permission to create and update customer support cases.
- Read-only access for AI Suggested Response.
- Editable access for fields such as Priority, Owner, and SLA Due Date.
- Access to the AI-Powered Customer Support application.

Support Manager Profile



The Support Manager profile was designed for supervisors responsible for monitoring support operations and team performance.

The profile includes:

- Full access to Case records.
- Access to Reports and Dashboards.
- Permission to monitor SLA performance.
- Ability to review AI-generated suggestions.
- Access to support analytics.

Activity 2: Create Custom fields for Customer Account

Permission Sets were created to grant additional permissions without modifying the user profiles.

Permission Set 1 – AI Feature Access Steps

1. Navigate to **Setup** → **Permission Sets**.
2. Click **New**.
3. Enter:
 - Label: **AI Feature Access**
 - User License: **None**
4. Save the Permission Set.
5. Configure the required object permissions.
6. Assign the Permission Set to users who require AI-related functionality.

Purpose

The AI Feature Access Permission Set enables users to access AI-related fields and functionalities such as:

- AI Suggested Reply
- AI Suggested Response
- AI Feedback

Permission Set	
AI Feature Access	
Find Settings... Clone Edit Properties Manage Assignments View Summary	
Permission Set Overview	
Description	API Name AI_Feature_Access
License	Namespace Prefix
Session Activation Required ☐	Created By Yarrabathula Naga Mounika, 7/3/2026, 9:20 PM
Permission Set Groups Added To 0	Last Modified By Yarrabathula Naga Mounika, 7/3/2026, 9:21 PM

Permission Set 2 – Report Access Steps

1. Navigate to **Setup** → **Permission Sets**.
2. Click **New**.
3. Enter:
 - Label: **Report Access**
4. Save the Permission Set.
5. Enable permissions for:
 - Create Reports
 - Customize Reports
 - Subscribe to Reports
6. Assign the Permission Set to the Support Manager profile.

Purpose

This Permission Set allows managers to create, customize, and monitor reports used for tracking support activities and overall system performance.

Permission Set

Report Access

Find Settings...CloneEdit PropertiesManage AssignmentsView Summary

Permission Set Overview

Description

License

Session Activation Required

Permission Set Groups Added To

API Name

Namespace Prefix

Created By

Last Modified By

Report_Access

Yarrabathula Naga Mounika7/3/2026, 10:05 PM

Yarrabathula Naga Mounika7/3/2026, 10:06 PM

Activity 3: Creating Users

Three users were created to simulate different roles within the customer support environment.

Steps

1. Navigate to Setup → Users.

2. Click New User.

3. Enter the required user details.

4. Assign the appropriate Profile.

5. Save the user record.

Users Created

User	Profile
Admin User	System Administrator
Support Agent	Support Agent
Support Manager	Support Manager

SETUPUsers

All Users

On this page you can create, view, and manage users.

To get more licenses, use the Your Account app. [Let's Go](#)

View: All UsersEditCreate New User

New UserReset Password(s)Add Multiple Users

Action	Full Name	Alias	Username	Role	Active	Profile
☐ Edit	Agent_Support	agent	supportagent2026@hello.com		✓	Support Agent
☐ Edit	Chatter Expert	Chatter	chatty.00dg500000d6bu9eaf.ay1xmft2hw1@chatter.salesforce.com		✓	Chatter Free User
☐ Edit	EinsteinServiceAgentUser	einstein	ai_customer_support_agent@00dg500000d6bu995707426.ext		✓	Einstein Agent User
☐ Edit	Manager_Support	smanager	manager996237@gmail.com		✓	Support Manager
☐ Edit	Naga Mounika Yarrabathula	mou	mounikan237.5e2d00beb61c@agentforce.com		✓	System Administrator
☐ Edit	User_Admin	admin	admin.cst12@demo.com		✓	System Administrator
☐ Edit	User_Integration	integ	integration@00dg500000d6bu9eaf.com		✓	Analytics Cloud Integration User
☐ Edit	User_Security	sec	insightssecurity@00dg500000d6bu9eaf.com		✓	Analytics Cloud Security User

New UserReset Password(s)Add Multiple Users

Milestone 7 : Page Layout Configuration

Activity 1: Configure Case Page Layout

The Case Page Layout was customized to display all important fields required for managing customer support requests. This ensures that support agents can easily access and update case information from a single interface.

Steps

1. Navigate to **Setup** → **Object Manager** → **Case**.
2. Select **Page Layouts**.
3. Open the default Case Page Layout.
4. Add the required custom fields to the layout:
 - Issue Type
 - Resolution Time
 - SLA Due Date
 - AI Suggested Reply
 - AI Suggested Response
 - AI Feedback
5. Arrange the fields in a logical order.
6. Save the layout.

Outcome

The customized page layout provides a centralized and user-friendly interface for viewing and updating customer support cases.

The screenshot displays the Salesforce 'Case' object page layout configuration. The left sidebar shows the navigation menu with 'Case Page Layouts' highlighted. The main area shows the 'Case' object with a list of fields and a 'Fields' section. The 'Fields' section includes a table with columns for field name, type, and status. Below this, there are sections for 'Quick Actions in the Salesforce Classic Publisher' and 'Salesforce Mobile and Lightning Experience Actions'. The 'Case Detail' section shows a list of fields and their values, including Case Owner, Case Number, Contact Name, and Case Reason. The 'Web Information' section shows fields for Web Email, Web Phone, and Web Name.

Field Name	Type	Status
Case Owner	Text	Active
Case Number	Text	Active
Contact Name	Text	Active
Case Reason	Text	Active
Issue Type	Text	Active
Resolution Time	Text	Active
SLA Due Date	Text	Active
AI Suggested Reply	Text	Active
AI Suggested Response	Text	Active
AI Feedback	Text	Active

Quick Actions in the Salesforce Classic Publisher

Post, Log a Call, Change Status, File, Link, Poll

Salesforce Mobile and Lightning Experience Actions

Actions in this section are predefined by Salesforce. You can [override the predefined actions](#) to set a customized list of actions on Lightning Experience and mobile app pages that use this layout. If you customize the actions in the Quick Actions in the Salesforce Classic Publisher section, and have saved the layout, then this section inherits that set of actions by default when you click to override.

Case Detail

Field Name	Value
Case Owner	Sample Text
Case Number	00N004201246-10124
Contact Name	Sample Text
Case Reason	Sample Text
Issue Type	Sample Text
Resolution Time	7/10/2018, 10:10 PM
SLA Due Date	7/10/2018, 10:10 PM
AI Suggested Reply	Sample Text
AI Suggested Response	Sample Text
AI Feedback	Sample Text

Web Information

Field Name	Value
Web Email	Sample Text
Web Phone	Sample Text
Web Name	Sample Text

Milestone 8 - Reports

To Differentiate the records and data stored in Customer Accounts, let's create two different Record Types.

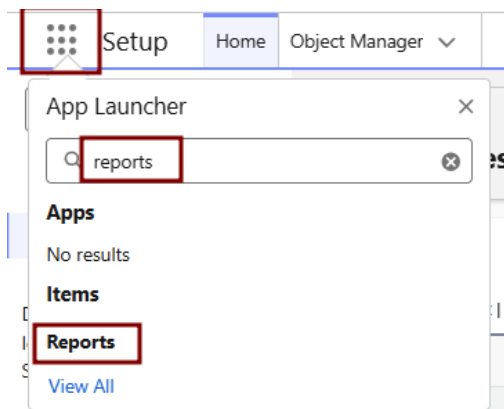
Online orders and Walk-In orders.

Activity 1: Create Customer Support Reports

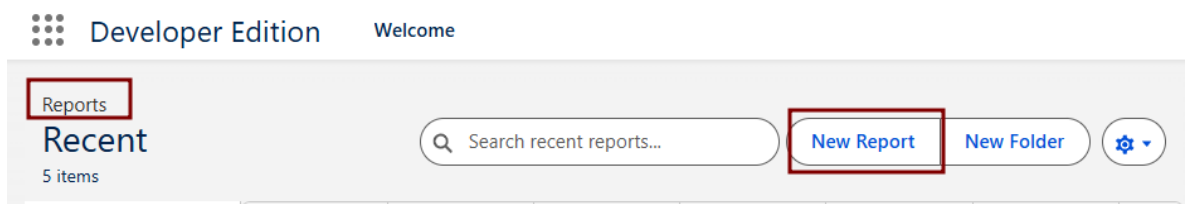
Reports were created to monitor customer support activities and evaluate overall support performance. These reports provide valuable insights into case distribution, priority levels, issue categories, and resolution performance.

Steps

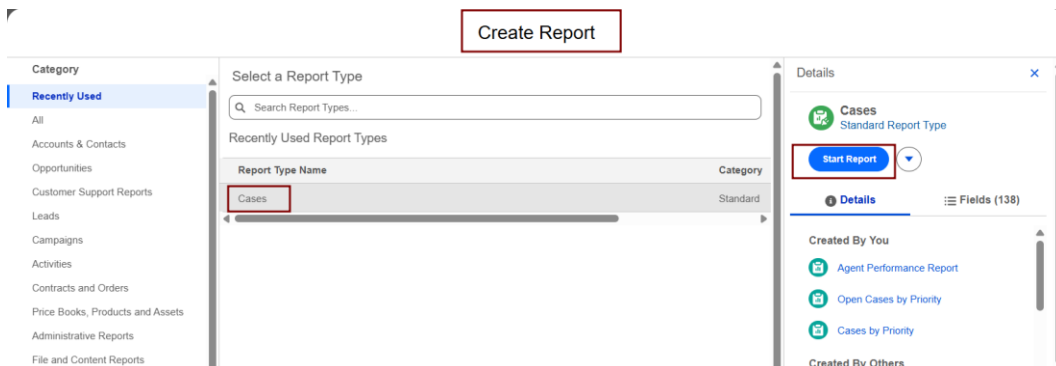
1. Navigate to **App Launcher** → **Reports**.



2. Click **New Report**.



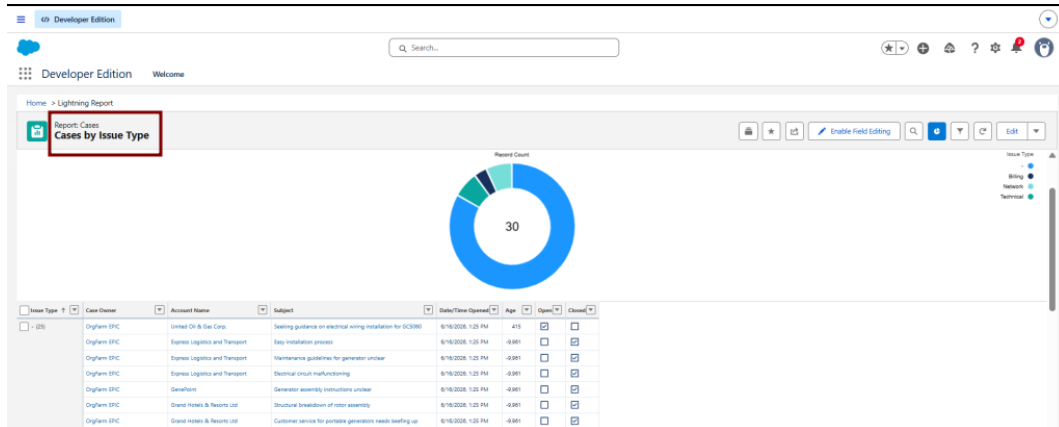
3. Select **Cases** as the report type.



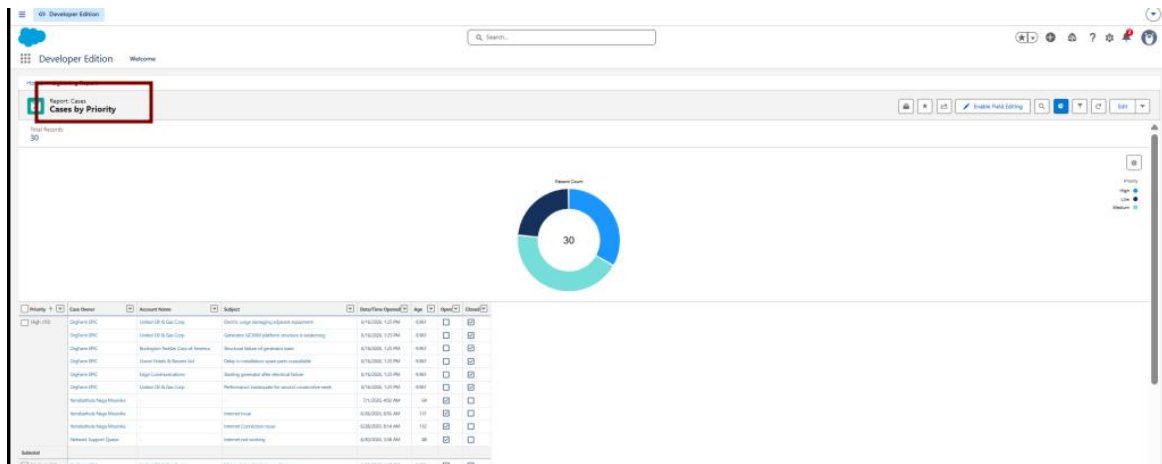
- Configure the required report filters and columns.
- Group the records based on the required field.
- Add charts wherever applicable.
- Save the report.

Reports Created

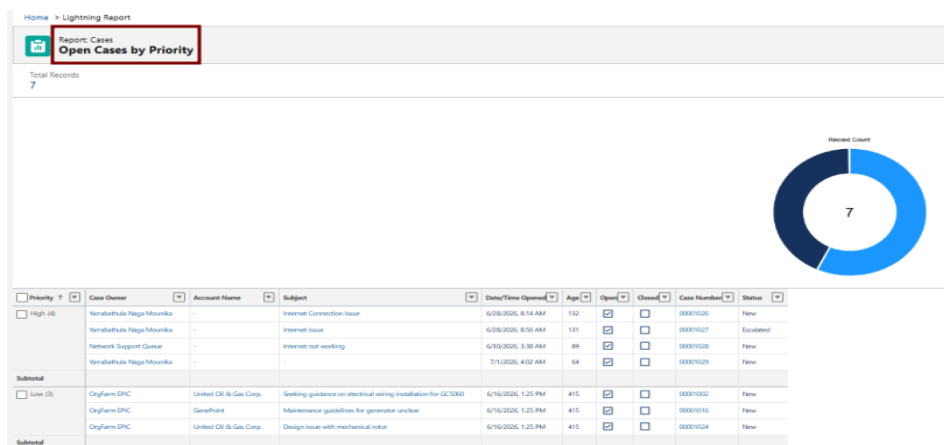
- Cases by Issue Type



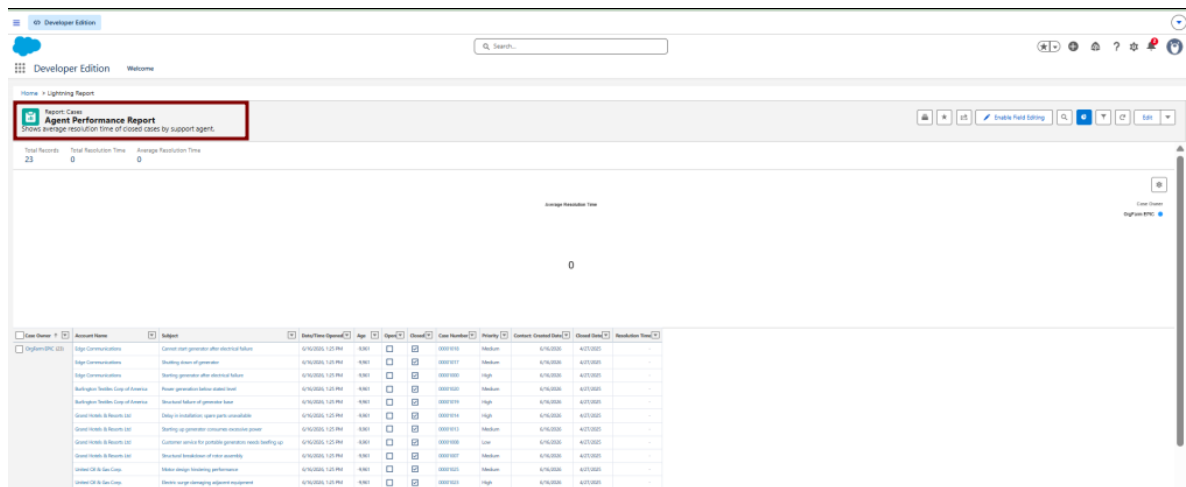
- Cases by Priority



- Open Cases by Priority



- Average Resolution Time



Outcome

The reports enable support managers to monitor support requests, identify high-priority issues, analyze case trends, and evaluate team performance. These reports help improve decision-making and ensure better customer service.

1. .

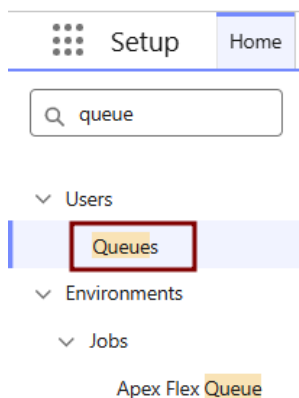
Milestone 9 – Queue

Activity 1: Configure Support Queues

Support queues were created to automatically organize and route customer support cases based on different issue categories.

Steps

1. Navigate to **Setup** → **Queues**.



2. Click **New Queue**.
3. Enter the Queue Name.

4. Select **Case** as the supported object.
5. Save the queue.
6. Repeat the process for all required support categories.

Queues Created

- Billing Support Queue
- Network Support Queue
- Hardware Support Queue
- Software Support Queue
- General Support Queue

Outcome

The queue mechanism enables efficient distribution of support cases to the appropriate support teams, reducing manual assignment and improving response time.

Queues

Queues allow groups of users to manage a shared workload more effectively. A queue is a location where records can be routed to await processing by a group member. The records remain in the queue until a user accepts them for processing or they are transferred to another queue. You can specify the set of objects that are supported by each queue, as well as the set of users that are allowed to retrieve records from the queue.

View: [All](#) | [Edit](#) | [Create New View](#)

Action	Label	Queue Name	Queue Email	Supported Objects	Modified By	Last Modified Date
Edit Del	Billing Support Queue	Billing_Support_Queue		Case	Naga Mounika Yarrabathula	6/30/2026, 3:21 AM
Edit Del	General Support Queue	General_Support_Queue		Case	Naga Mounika Yarrabathula	6/30/2026, 3:19 AM
Edit Del	Hardware Support Queue	Hardware_Support_Queue		Case	Naga Mounika Yarrabathula	6/30/2026, 3:20 AM
Edit Del	Network Support Queue	Network_Support_Queue		Case	Naga Mounika Yarrabathula	6/30/2026, 3:20 AM
Edit Del	Software Support Queue	Software_Support_Queue		Case	Naga Mounika Yarrabathula	6/30/2026, 3:20 AM

Milestone 10 - System Testing

Activity 1: Functional Testing

After implementing all project components, functional testing was performed to verify that each feature worked correctly.

Testing Performed

- Verified successful creation and updating of Case records.
- Tested Validation Rules to ensure mandatory fields were enforced.
- Verified automatic execution of Record-Triggered Flows.
- Confirmed automatic case routing based on Issue Type.
- Tested user access using different Profiles and Permission Sets.
- Verified reports generated accurate information.

Outcome

The system functioned successfully, with all automation and security features working as expected. Validation rules prevented incorrect data entry, flows automated case processing, and reports displayed accurate support information.

The screenshot displays a Salesforce Case record form. The form includes fields for Case Number, Contact Name, Account Name, Type, Case Reason, Issue Type, SLA Due Date (with Date and Time sub-fields), Resolution Time, and a checkbox for 'Send notification email to contact'. The 'Priority' dropdown is set to 'High', and the 'Case Origin' dropdown is set to 'Phone'. A red box highlights the 'SLA Due Date' field, and a red arrow points to the validation message: 'SLA Due Date must be populated for High priority cases.' A modal dialog box is open, displaying the message 'We hit a snag.' and a list of fields to review, which includes 'SLA Due Date'. The 'Save' button is highlighted in blue.

Developer Edition

Search...

Case: Internet not working

Follow Edit Delete Change Owner

Feed Related

Post Poll

Share an update...

Share

Most Recent Activity

Search this feed...

All Updates Call Logs Text Posts Status Changes

Yarrabathula Naga Mounika Case updated

Case Owner (using assignment rule): Yarrabathula Naga Mounika to Network Support Queue

1m ago

Details

Case Owner	Network Support Queue	Status	New
Case Number	00001028	Priority	High
Contact Name	Naga Mounika Yarrabathula	Contact Phone	
Account Name		Contact Email	
Type		Case Origin	
Case Reason		Email	
Issue Type	Network		
SLA Due Date	6/30/2026, 12:00 PM		
Resolution Time			

Gmail

Search mail

Compose

Inbox 3,906

Starred

Snoozed

Sent

Drafts 15

Purchases 21

More

Labels +

Case transferred to you.

Case Notification <noreply@salesforce.com> to me

Case 00001028 has been assigned to you. Please click on the link below to view the record.

<https://orgfarm-d8a7bd4d1b-dev-ed.develop.my.salesforce.com/500g500000ePvmO>

16:08 (4 minutes ago)

Reply Forward

Case Information

Case Owner

 Yarrabathula Naga Mounika

* Status

New

Case Number

Priority

High

Contact Name

 Naga Mounika Yarrabathula

* Case Origin

Email

Account Name

Search Accounts...

Type

--None--

Case Reason

--None--

Issue Type

--None--

Please select an Issue Type.

SLA Due Date

Date: 7/1/2026 Time: 12:00 PM

Resolution Time

Support Category

☐ Assign using active assignment rule

AI Feedback

☐ Send notification email to contact

Search AI Feedbacks...

⚠ We hit a snag.

Review the following fields

- [Issue Type](#)



Cancel

Save & New

Save

Milestone 12 - Conclusion

The AI-Powered Customer Support Ticketing System was successfully developed using Salesforce Developer Edition. The system centralizes customer support operations by managing Accounts, Contacts, and Cases while automating case classification, priority assignment, and SLA tracking through Salesforce Flows. Profiles, Permission Sets, Users, and Queues provide secure role-based access, ensuring that users perform only their authorized tasks. Reports enable support managers to monitor case trends and overall support performance.

Overall, the project demonstrates how Salesforce's low-code platform can be used to build a secure, scalable, and automation-driven customer support solution that improves operational efficiency, reduces manual effort, and enhances customer satisfaction.

7. FUNCTIONAL & PERFORMANCE TESTING

The application was tested thoroughly using Salesforce Lightning Experience to ensure that all implemented features functioned correctly.

Functional Testing

- Case creation and update
- Custom field verification
- Validation Rule testing
- Record-Triggered Flow execution
- Profile and Permission Set verification
- Queue configuration
- Report generation

Performance Testing

Performance testing confirmed that the application responded quickly while processing customer support cases. Reports loaded efficiently, Flows executed automatically without noticeable delay, and user access remained stable throughout testing.

8. ADVANTAGES & LIMITATIONS

Advantages

- Centralized customer support management

- Automated case classification and priority assignment
- Secure role-based access control
- Reduced manual effort through Salesforce Flows
- Improved reporting and support monitoring
- Scalable cloud-based architecture
- AI-assisted customer support features

Limitations

- Requires a Salesforce Developer or licensed environment
- Internet connectivity is required
- AI suggestions depend on configured automation rather than advanced machine learning
- Dashboards require additional customization for advanced analytics

9. Conclusion:

The AI-Powered Customer Support Ticketing System successfully streamlines customer service operations by integrating Salesforce standard objects, automation, security, and reporting features into a single platform. The implementation improves case management, enhances operational efficiency, ensures secure access to data, and supports better decision-making through automated workflows and reports. The project provides a strong foundation for building intelligent customer support solutions using Salesforce.

10. FUTURE SCOPE

Future enhancements that can further improve the system include:

- Integration with Salesforce Einstein AI
- Agentforce-powered intelligent support agents
- Omni-Channel support for multiple communication platforms
- AI-based case prediction and automatic recommendations
- Chatbot integration for customer self-service
- Mobile application support
- Knowledge Base integration
- Real-time SLA monitoring and predictive analytics

11. APPENDIX

Project Components

- Salesforce Developer Edition
- Standard Objects (Account, Contact, Case)
- Custom Object (AI Feedback)
- Custom Fields
- Validation Rules
- Record-Triggered Flows
- Profiles
- Permission Sets
- Users
- Queues
- Reports

Tools Used

- Salesforce Developer Edition
- Salesforce Lightning Experience
- Flow Builder
- Reports
- Object Manager

Thank You